

Analytical Results and Policy-Relevant Findings

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1 Introduction

This note presents the results of analytical applications derived from the natural capital accounts (NCA) reviewed in the companion diagnostic report. The analysis draws on compiled SEEA accounts for the Democratic Republic of Congo, Ethiopia, Kenya, Madagascar, Mauritius, Rwanda, South Africa, Uganda, and Zambia, supplemented where noted by online data and literature on the World Bank operational programs to which the NCA data most directly connects.

The organizing principle here is thematic rather than country-by-country, where each section addresses one of the application areas, with country-level evidence presented in country alphabetical order under each theme.

2 Energy and Minerals: Informing the ASCENT Program

The World Bank’s Accelerating Sustainable and Clean Energy Access Transformation (ASCENT) program, launched jointly with COMESA in Lusaka in June 2024, represents the largest coordinated energy investment in Eastern and Southern Africa’s history. Its target is to deliver new electricity connections to 100 million people across up to 21 AFE countries by 2030, through a USD 5 billion IDA envelope complemented by an additional USD 10 billion from development partners, climate financiers, and the private sector. The program operates across three pillars: regional and national platforms for harmonized planning and regulation; grid densification and renewable integration; and scaling of distributed renewable energy (DRE) and clean cooking technologies. In March 2026, the Green Climate Fund approved a USD 250 million ASCENT-GREEN sub-program, targeting 21 million new electricity connections, clean cooking for 12 million people, and productive energy use for 5 million farmers and SMEs across 21 countries including all countries reviewed in this note. The energy and mineral accounts compiled by countries in the AFE region provide baseline data directly relevant to ASCENT’s design and monitoring.

Democratic Republic of Congo. The DRC’s forest ecosystem accounts establish that 94 percent of the country’s total energy supply comes from forest biomass (firewood and charcoal combined) and 98.8 percent of household energy is biomass-based (Turpie et al., 2025). Only approximately 10 percent of the population has reliable electricity access, making the DRC one of the most acute energy poverty contexts in the ASCENT portfolio. The accounts further document that fuelwood harvesting from forest-farm mosaics increased 709 percent between 2000 and 2020, which represents a structural change driven by both population growth and the absence of affordable electricity alternatives. For ASCENT, this creates a compelling dual-entry point, because clean cooking programs can simultaneously address household energy access and reduce one of the primary drivers of forest degradation documented in the accounts. Carbon finance mechanisms aligned with Article 6 of the Paris Agreement represent a potential additional revenue stream, given that DRC forests store approximately 204,000 million tonnes of CO₂ equivalent (Turpie et al., 2025).

Ethiopia. Ethiopia’s accounts do not yet include compiled energy supply and use tables, but the ecosystem services accounts document that over 90 percent of Ethiopian households rely on wood as their primary energy source, representing a very large but unmeasured contribution to household energy demand. The accounts indicate that shrubland conversion to cropland is generating net carbon emissions, with open shrubland alone emitting 194.8 million tonnes C per year nationally (Rutebuka et al., 2025). Ethiopia is among the 21 countries covered by ASCENT-GREEN, and its participation in carbon markets under Article 6, explicitly noted in the land accounts, could generate revenue to offset the cost of energy transition programs. A future energy supply and use table for Ethiopia would provide the physical baseline needed to design ASCENT interventions and monitor their impact on biomass demand.

Kenya. Kenya’s monetary energy supply and use table (MSUT), which is the first in the AFE region, provides a model for the kind of account most directly relevant to ASCENT

planning (Kenya National Bureau of Statistics, 2026a; World Bank & Prime Africa Consult, 2025). The MSUT reveals that, while biomass dominates the physical energy balance (over 1.2 million TJ in household use), households concentrate their monetary energy spending on electricity and petroleum products, making these the priority sectors for energy transition finance. Kenya’s electricity access has climbed to approximately 76 percent, the highest in the AFE group, and the accounts document the structure of the energy supply chain in monetary terms, enabling analysis of where ASCENT investments would generate the greatest multiplier effects. [inventory item: KEN.energy.1 about here]

Uganda. Uganda’s forest accounts establish that woody biomass supplies 78 percent of national energy (Uganda Bureau of Statistics, 2020). Charcoal production grew from 2,396 thousand tonnes in 1990 to 11,962 thousand tonnes in 2015 (a fivefold increase) at a kiln efficiency of only 12.5 percent, consuming 95.7 million tonnes of wood per year. The accounts projected in 2020 that sustainable wood supply would be fully depleted by approximately 2025 under business-as-usual conditions, with a supply deficit growing to 105 million tonnes by 2040 (Uganda Bureau of Statistics, 2020). Uganda’s oil sector water abstraction grew 113.9 percent year-on-year in 2024 (Uganda Bureau of Statistics, 2025), signalling the emerging demands of the oil development program on both water and, by extension, energy infrastructure. For ASCENT, Uganda’s deforestation trajectory driven by charcoal demand represents one of the clearest cases where clean cooking scale-up would generate co-benefits across energy access, forest conservation, and climate.

South Africa. South Africa’s energy flow accounts, which are the most detailed in the region, quantify both the structure and vulnerabilities of the national energy system (Statistics South Africa, 2025). Coal accounts for 72.2 percent of domestic natural energy inputs and generates 90.4 percent of energy exports (2,046.9 PJ in 2022). Despite coal self-sufficiency, overall import dependency rose to 19.9 percent (2022), with oil at 93.1 percent import dependency and natural gas at 99.9 percent. South Africa is an IBRD-eligible country participating in ASCENT through IBRD borrowing, primarily for grid strengthening and renewable energy integration. The energy accounts provide the quantified baseline for assessing ASCENT’s impact on South Africa’s energy dependency ratio, export revenues, and household energy costs. [inventory item: ZAF.energy.1 about here] [inventory item: ZAF.energy.2 about here]

Zambia. Zambia’s accounts document the same structural energy pattern as Uganda, where charcoal and fuelwood account for 78 percent of total national energy consumed, creating a direct structural linkage between deforestation and household energy security (Chisanga et al., 2026). Zambia’s Lake Kariba fell below 15 percent of capacity in both 2016 and 2024, in each case triggering severe nationwide electricity load-shedding and costly electricity imports. The 2023/24 drought that was declared a national disaster simultaneously crippled hydropower generation and destroyed the maize crop. For ASCENT’s third pillar (scaling distributed renewables), Zambia represents a priority case, because it is a country where hydropower dependency combined with biomass reliance creates a double vulnerability to climate variability, and where solar and clean cooking solutions address both exposure pathways simultaneously.

3 Forests: Contribution to GDP and the MBJ Indicator

The World Bank’s More and Better-Paid Jobs (MBJ) agenda frames job creation as a central development outcome alongside economic growth. Estimating the employment elasticities of forest sector growth (i.e. the number of additional jobs generated per unit increase in forest value added) is a prerequisite for using NCA data to inform the MBJ indicator. The World Bank’s pilot work on indirect jobs estimation uses Social Accounting Matrix (SAM) models and employment-to-value-added elasticities to compute both direct and indirect employment multipliers from sectoral investments. In the forest sector, the primary challenge is that formal sector employment represents a small fraction of actual forest-dependent livelihoods. In Uganda, for example, only 4.7 percent of the monetary value of wood available for supply was captured in the formal economy in 1990 (Uganda Bureau of Statistics, 2020). For instance, in newly approved programs, the World Bank usually projects direct forest jobs. However, these projections are based on assumptions about value chain formalization and smallholder processing capacity rather than on country-specific NCA employment elasticity data. The accounts reviewed here provide the physical and monetary baselines from which sector-specific elasticity estimates could be derived, particularly where monetary forest asset accounts or monetary ecosystem services accounts are available.

Democratic Republic of Congo. The DRC’s forest accounts provide the most comprehensive forest sector data in the region (Turpie et al., 2025). The forestry sector contributed approximately 9 percent of GDP in 2021 (down from 11 percent in 1990), with primary processed wood product exports of approximately USD 141 million in 2022. Informal and small-scale logging accounted for nearly 90 percent of logging activity in the DRC in 2014, meaning that official GDP statistics substantially underestimate the true economic contribution of the forest sector. Bushmeat income represents approximately 32 percent of annual household income in some forest-dependent communities. The forest asset value increased from USD 5.4 trillion (2000) to USD 10.3 trillion (2020) in total international asset value terms, but per capita forest wealth declined over the same period as population grew faster than sustainable forest income. Forest depletion increased 113 percent between 2010 and 2020, indicating that nominal asset growth masks accelerating unsustainable extraction. For the MBJ indicator, the accounts provide sectoral GDP contributions and physical output volumes from which employment elasticities could be calibrated, particularly for non-timber forest products and charcoal value chains. [inventory item: DRC.forest.1 about here] [inventory item: DRC.forest.3 about here]

Ethiopia. Ethiopia’s forest sector is estimated to contribute approximately 13 percent of GDP when informal and subsistence uses are included, or around 6 percent on a formal sector basis (MEFCC, 2017). In the 2012-13 period, Ethiopian forests generated cash and in-kind economic benefits equivalent to USD 16.7 billion (UNEP, 2016). Coffee, grown predominantly in forest systems in southwest Ethiopia, generates USD 1.4 billion in annual export earnings, representing 35 percent of total export revenue. The ecosystem services accounts document that forest and cropland ecosystems sequester carbon at scale, suggesting that REDD+ and Article 6 mechanisms represent a potential employment-generating revenue stream for restoration

programs under ESIF-II. Physical accounts are in place; a monetary ecosystem services account would be the next step required to translate ecosystem service flows into sector-level value added estimates usable in MBJ calculations.

Kenya. Kenya’s forest accounts (Kenya National Bureau of Statistics, 2026b) note that the Plantation Establishment and Livelihood Improvement Scheme (PELIS), which allocates forest-adjacent land to communities for intercropping, generates an estimated KSh 50,000 per hectare per year in measurable community benefit. The KENMOD macroeconomic and fiscal model integration enables forest-related scenarios to be translated directly into projections for GDP, fiscal revenues, and employment, making Kenya the only country in the AFE region with a functional tool for estimating MBJ-type employment-to-GDP linkages for the forest sector. Forest cover declined from 6.65 to 6.07 percent (1990–2024), with formal forest accounts noting gaps in informal timber flows and non-timber forest products. [inventory item: KEN.forest.1 about here]

Uganda. Uganda’s monetary timber asset accounts provide the most detailed monetary forest sector data in the AFE region after the DRC (Uganda Bureau of Statistics, 2020). Total forest land value was USD 34.9 billion in 2015, despite a 60 percent loss in forest area since 1990, reflecting rapid escalation in per-hectare land values, especially in the Central region (USD 129,700/ha). The monetary wood asset stock declined from USD 9.1 billion (1990) to USD 7.4 billion (2015) at stumpage prices. Wood product supply reached USD 270.6 million in 2015, with sawn timber (USD 111.7M) and poles (USD 77.7M) as the largest categories. Non-wood forest product (NWFP) monetary accounts show *Prunus africana* bark at UGX 265,195 million and shea butter at UGX 14,553 million. For MBJ estimation, Uganda’s accounts provide the value-added by product category required to calibrate employment elasticities across formal and informal forest sub-sectors. Forestry contributes between 4 and 8 percent of GDP when informal uses are included. [inventory item: UGA.forest.1 about here] [inventory item: UGA.forest.2 about here] [inventory item: UGA.forest.4 about here]

Zambia. Zambia’s miombo woodland accounts document both the economic scale and the structural vulnerability of forest-dependent livelihoods (Chisanga et al., 2026). The accounts record 109,968 tonnes of wild fruits, 83,177 tonnes of leaves, 74,454 tonnes of mushrooms, 44,842 tonnes of tubers, and 2,097 tonnes of honey harvested annually. Charcoal, which drives the dominant conversion from trees to rangeland (4.9 million ha between 2018-2023), represents an informal economy whose employment significance far exceeds its monetary capture in formal accounts. An estimated 60 percent of Zambians rely on medicinal plants, indicating a very large but unquantified natural capital contribution to health and welfare. [inventory item: ZMB.wildlife.3 about here]

4 Water: Contribution to Agriculture, Industry, and Manufacturing

Water accounts in the AFE region demonstrate most clearly the role of natural capital accounting as a bridge between environmental monitoring and economic policy. The pattern across countries

is consistent, showing that water is simultaneously the most constrained production input, the most socially critical resource, and the natural capital asset with the clearest macroeconomic linkages (to agriculture, hydropower, industrial production, and sanitation). Countries that have compiled monetary water accounts (Rwanda and, partially, Uganda) are able to make the specific economic arguments for water allocation reform, infrastructure investment, and pricing policy that physical accounts alone cannot support.

Kenya. Kenya’s water accounts reveal that per capita water availability varies by a factor of eleven across basins, from 8 m³ per person per year in Lake Victoria South to 91 m³ in the Tana basin, while the Rift Valley shows groundwater abstraction exceeding 100 percent of sustainable yield (World Bank & Prime Africa Consult, 2025). Electricity generation is the largest single sectoral water user, establishing the water-energy nexus as a direct planning concern. The accounts document that non-revenue water declined modestly, and explicitly propose a SEEA water quality accounting structure for future compilation. Kenya’s monetary ecosystem services accounts separately value sediment retention services protecting downstream reservoir capacity at KSh 23,912 million per year (2018), with sedimentation rates at the Tana hydropower cascade reaching 1.45 percent per year, which is a direct monetary argument for catchment conservation investment. [inventory item: KEN.water.1 about here] [inventory item: KEN.water.2 about here]

Mauritius. Mauritius provides the most institutionalized water accounting system in the region, in its fifth consecutive edition (Statistics Mauritius, 2024). The accounts reveal that while Mauritius remains below the 25 percent water stress threshold nationally, agriculture accounts for 89.1 percent of net water consumption, which is a structural dominance that is rarely visible in sector-level GDP data. The most policy-significant finding is the worsening unaccounted-for water (UFW) rate, which went up from 56.4 percent in 2013 to 64.6 percent in 2022, meaning nearly two-thirds of treated potable water abstracted by the Central Water Authority is lost before reaching end users. Households received only 197 litres per capita per day despite 99.8 percent access to safely managed drinking water services, because the gap between abstraction (719 L/day per capita) and delivery is almost entirely explained by distribution infrastructure deterioration. A monetary water supply and use table that quantified the treatment cost and foregone utility revenue of UFW would provide a direct economic argument for pipe network investment. [inventory item: MUS.water.3 about here]

Rwanda. Rwanda’s monetary water supply and use tables, one of only two in Africa, provide the most direct link between water use and economic productivity available in the region (Government of Rwanda, 2019b). National water use efficiency reached 4,762 Rwf per m³ in 2015, up from approximately 4,500 Rwf/m³ in 2012. The sector breakdown reveals that agriculture consumes over 96 percent of total freshwater abstraction but achieves water productivity of only 118 Rwf/m³, compared to 523 Rwf/m³ for manufacturing, 6,236 Rwf/m³ for mining, and 9,300–11,600 Rwf/m³ for services. Rwanda’s GDP grew 25 percent between 2012 and 2015 while water use grew only 10 percent, demonstrating relative decoupling, which is the kind of macroeconomic efficiency indication that justifies water pricing reform and investment in irrigated agriculture. Climate impacts on water resources are projected to cost approximately 1

percent of GDP per year by 2030. WASAC's non-revenue water of 41 percent represents USD 8.7 million in annual foregone revenue, which is a figure that is directly quantifiable in monetary water accounts. [inventory item: RWA.water.1 about here] [inventory item: RWA.water.2 about here]

Uganda. Uganda's water accounts confirm the dominant role of rainfed agriculture (56 percent of total abstraction) and the low uptake of irrigation (0.5 percent of agronomic potential) (Uganda Bureau of Statistics, 2025). Water productivity reached UGX 5,557 per m³ consumed in 2024, a growth of 84 percent above the 2015 baseline, but with agriculture remaining the lowest-productivity sector (UGX 1,458/m³) compared to services (UGX 364,299/m³). The accounts also flag an emerging pressure point, because the oil sector water abstraction grew 113.9 percent year-on-year in 2024, from 164,676 to 352,188 thousand m³, showing the demands of Uganda's oil development programme. Distribution losses of 34.8 percent in the water supply network indicate substantial infrastructure investment needs. [inventory item: UGA.water.3 about here]

South Africa. South Africa's water accounts focus on Strategic Water Source Areas (SWSAs), which cover only 8.2 percent of national land area but supply approximately two-thirds of national economic activity and over 90 percent of urban water users (Statistics South Africa, 2023). Three SWSAs, consisting of Upper Usutu (40.9 percent natural), Table Mountain (50.4 percent), and Mpumalanga Drakensberg (51.3 percent), have already fallen below the 60 percent natural land cover threshold considered necessary for functional water production, threatening water supply to Gauteng, which hosts the country's economic core. The SWSA accounts document that commercially irrigated pivots expanded 221.4 percent nationally (1990–2014), with the largest expansion in the already water-scarce Free State (+489.7 percent), creating an acute water demand growth in the catchments least able to absorb it. The monetary integration of SWSA ecosystem accounts with national water accounts would enable direct cost-benefit comparison between catchment conservation investments and alternative water supply infrastructure. [inventory item: ZAF.water.4 about here] [inventory item: ZAF.water.2 about here]

Zambia. Zambia's water accounts situate water not only as a production input but as a source of macroeconomic vulnerability (Chisanga et al., 2026). The country's structural exposure is dual, because hydropower at Lake Kariba, Kafue Gorge, and Itzhi-Tezhi underpins electricity supply, while agriculture is the largest consumptive user. Irrigation for wheat, sugarcane, and export crops competes directly with hydropower in the Kafue Basin. The 2023/24 drought, the most severe in recent history, illustrated this dual exposure, because maize crops failed in southern and central regions, Lake Kariba fell below 15 percent of capacity, and the government was forced to import electricity at substantial cost. Bangweulu wetlands alone generate an estimated US\$11.7 million per year in ecosystem services, supporting approximately 490 freshwater fish species including 138 endemic species in Lake Tanganyika. Water quality hotspots in the Copperbelt (heavy metals from acid mine drainage) and Lusaka (pathogen contamination contributing to cholera outbreaks) represent risks that are documented in the accounts but not yet monetized.

5 Additional Applications

Beyond the three primary thematic areas, the reviewed accounts yield a set of additional policy-relevant applications that address biodiversity, climate finance, sustainable fisheries, and landscape investment targeting. These applications span both monetary and physical account types and illustrate the breadth of potential uses for NCA data in operations.

Ethiopia. Ethiopia’s ecosystem services accounts provide quantified physical evidence for watershed investment decisions that could be directly applied in ESIF-II program design (Rutebuka et al., 2025). Ecosystems avoided an estimated 29.7 billion tonnes of soil erosion in 2022, but the service delivered per hectare of cropland fell 18 percent (from 411 to 338 t/ha) between 2013 and 2022, which shows that landscape conversion is reducing the protective cover of surrounding ecosystems even as agricultural area expands. The Abbay basin accounts for over 40 percent of national soil protection value. Dry-season baseflow declined 1.3 percent and flood mitigation capacity declined 2.7 percent, while Addis Ababa records a negative flood mitigation value of $-2,691 \text{ m}^3/\text{ha}$, meaning the city amplifies rather than attenuates peak flows. Carbon sequestration services of forests and croplands represent a potential Article 6 revenue stream, but the net shrubland emission (from conversion) is eroding this value. [inventory item: ETH.ecosystem.2 about here] [inventory item: ETH.ecosystem.3 about here]

Madagascar. Madagascar’s landscape assessment documents that 35.6 percent of national territory (21.1 million hectares) is degrading, with 1.96 billion tonnes of carbon (31 percent of the national total) concentrated in the highest-priority NBS zones, which represent only 7.5 percent of national territory (Vogl et al., 2022). At the social cost of carbon (USD 38.1/tonne CO₂ in 2015, rising 3 percent annually), the annual carbon absorption loss from land degradation reaches approximately USD 59 million per year. The three main hydropower plants serving Antananarivo’s grid lose approximately 19 percent of installed generation capacity to watershed degradation. Implementing integrated landscape management in priority zones could reduce soil erosion by 81 percent and surface runoff by 73.5 percent within those areas. These physical metrics provide the foundation for a carbon finance and watershed protection investment program.

Rwanda. Rwanda’s ecosystem accounts quantify the biophysical cost of agricultural expansion in one of the world’s most densely populated rural landscapes (Government of Rwanda, 2019a). Soil erosion increased 54 percent between 1990 and 2015 (from 102.5 to 158.2 million tonnes per year), while sediment export to waterways increased 123 percent over the same period, with reservoir sediment loads growing 39 percent in just the five years from 2010 to 2015. Baseflow declined in every district, with per-household water availability already below basic-needs thresholds in several peri-urban Kigali districts. Wetlands declined from 110,000 ha to 88,773 ha between 1990 and 2015, with 14,000 ha lost in just 2010–2015. Soil erosion costs are estimated at US\$34 million per year (approximately 2 percent of GDP). These accounts provide direct evidence for the economic case for Rwanda’s terracing and agroforestry investment programs. [inventory item: RWA.ecosystem.1 about here] [inventory item: RWA.ecosystem.3 about here]

South Africa. South Africa’s biodiversity tourism accounts are the most complete monetary ecosystem services accounts for any sector in the AFE region (Statistics South Africa, 2024). Total biodiversity-based tourism generated R60,663 million in spending in 2019, with biodiversity-specific tourism GDP (BTDGDP) of R27,726 million (between 0.4 and 0.5 percent of GDP). Total tourism employment reached 780,096 jobs (4.8 percent of employment), of which 91,836 were directly attributable to biodiversity-based activities, where avitourism alone generates R2.24 billion per year. The estuarine ecosystem accounts show severe degradation, with only 4 percent of assessed estuaries showing good fish condition. Sawfish are now extinct, and dusky kob spawner biomass is below 2 percent of pristine levels (Niekerk et al., 2020). Together, these accounts make a quantified economic argument for protected area expansion, which grew from 7.3 to 9.2 percent of national territory between 2000 and 2020, as a return-generating investment in biodiversity assets. [inventory item: ZAF.biodiversity.1 about here] [inventory item: ZAF.biodiversity.2 about here]

Zambia. Zambia’s wildlife and fisheries accounts establish the economic contribution of natural resources to an economy where approximately 40 percent of national income is estimated to derive from natural capital (Chisanga et al., 2026). Fish production increased 42.4 percent between 2018 and 2023 (from 125,300 to 178,452 metric tonnes), with aquaculture growing 112.2 percent and total fish sector value reaching ZMW 8 billion (US\$422.6 million) in 2023. Bush meat revenue from Game Management Areas grew 158.5 percent in nominal terms, partly reflecting kwacha depreciation but also genuine volume growth (+41 percent). The net present value of carbon stock loss due to ecosystem degradation was estimated at USD 46 million (2018–2023), providing a monetary entry point for carbon finance discussions even in the absence of full monetary ecosystem accounts. [inventory item: ZMB.wildlife.1 about here] [inventory item: ZMB.wildlife.2 about here] [inventory item: ZMB.ecosystem.1 about here]

6 Cross-Country Comparisons and Trends

The accounts, as a whole, reveal regional patterns that are not visible in any single country’s data. Four cross-cutting findings stand out.

The charcoal/biomass parallel. Uganda and Zambia independently report that 78 percent of national energy comes from woody biomass (Chisanga et al., 2026; Uganda Bureau of Statistics, 2020). The DRC is at 94 percent biomass (Turpie et al., 2025). Ethiopia and Madagascar have equivalent structural dependencies but without compiled energy accounts. This is not a transitional feature of a few lagging countries. It characterizes the majority of AFE economies. The policy implication for ASCENT is that clean cooking scale-up is simultaneously an energy access intervention, a forest conservation measure, and a climate mitigation action. The accounts provide the physical basis for quantifying each dimension.

Forest loss trajectories. Uganda recorded a 60 percent loss of forest cover between 1990 and 2015; a rate of approximately 119,000 ha per year for 25 years (Uganda Bureau of Statistics, 2020). Zambia lost 1,882,000 ha of tree cover in just five years (2018-2023), a rate of 376,000

ha per year (Chisanga et al., 2026). Rwanda lost 673,000 ha of sparse forest between 1990 and 2015 (Government of Rwanda, 2018). The DRC lost 11.6 million hectares of forest cover overall between 2000 and 2020, with the deforestation rate more than doubling from 0.5 percent per year to 0.8 percent per year between the two decades (Turpie et al., 2025). Ethiopia, by contrast, reports a 22 percent increase in total forest cover over 2013-2022, though the accounts themselves caution that classification differences may explain this apparent gain. Taken together, the accounts document a region in active forest transition, where the pace of loss in some countries is outstripping the regeneration documented in others. [inventory item: ETH.land.1 about here] [inventory item: ZMB.land.1 about here]

Monetary valuation frontier. The region divides sharply between countries with monetary ecosystem service accounts and those without. Kenya and Rwanda have compiled monetary ecosystem services accounts. For instance, Kenya for livestock, sediment retention, and ecosystem-based tourism and Rwanda for water productivity by sector. South Africa has monetary biodiversity tourism accounts. Zambia has a single monetary estimate (carbon Net Present Value). Ethiopia, Madagascar, and the DRC have physical accounts only, despite substantial investment in biophysical modelling. This frontier matters because monetary accounts are what enable direct comparison with GDP, infrastructure costs, and fiscal revenue, which is essentially the language of economic planning. Cross-country analytical products would benefit significantly from harmonizing valuation approaches, particularly for carbon and water regulation, to enable meaningful regional comparison.

Water productivity gap. Where water productivity data are available, they reveal a consistent structural finding showing that agriculture dominates water abstraction but generates the lowest value per cubic metre. Rwanda’s agricultural water productivity (118 Rwf/m³) is 53 times lower than mining and nearly 80 times lower than services (Government of Rwanda, 2019b). Uganda’s agriculture (UGX 1,458/m³) is more than 250 times less productive per cubic metre than the services sector (UGX 364,299/m³) (Uganda Bureau of Statistics, 2025). This gap is not an argument against agricultural water use. It merely reflects that agriculture provides food security and employs the majority of the population. However, it does provide a quantified basis for designing water allocation reforms, promoting water use efficiency in irrigated agriculture, and pricing water to incentivize conservation.

7 Policy Relevance, Reflections, and Conclusions

7.1 Policy-Relevant Insights

Natural capital accounts can justify and target climate investments. Ethiopia’s soil protection accounts establish that degradation reduces agricultural GDP by between 2 and 6.75 percent annually. Rwanda’s accounts document a tenfold increase in reservoir sedimentation that is reducing hydropower capacity and water supply security. Zambia’s drought-driven electricity and food crisis of 2023/24 was a predictable outcome of the water account’s hydropower-drought

exposure indicators. These physical accounts, even where monetary valuation is absent, provide the evidence base for economic analysis of climate adaptation investments.

Energy accounts reveal transition risks and opportunities. Kenya’s monetary energy accounts expose that biomass dominance in the physical balance coexists with electricity and petroleum dominance in household monetary spending, which is a finding that changes the design logic for energy transition support. South Africa’s energy accounts quantify the country’s structural oil import vulnerability (93.1 percent import dependency), providing a macroeconomic case for accelerating renewable energy deployment. Uganda and Zambia’s accounts establish that charcoal demand is on a trajectory to exhaust available sustainable supply, creating a climate tipping point for both energy security and forest conservation that the accounts can track.

Forest accounts provide a foundation for forest-economy programs. The DRC’s accounts establish that the forestry sector contributes 9 percent of GDP but that informal activity accounts for nearly 90 percent of logging. This ratio provides the baseline for formalization programs, and for estimating the MJB employment multipliers that such programs are expected to generate. Uganda’s accounts provide the only sub-regional monetary forest asset data in the region and demonstrate that forest land values are escalating even as area is lost, an important dynamic for designing community land security programs.

Water accounts underpin investment targeting and pricing reform. Rwanda’s monetary water productivity accounts provide the most operationally useful data in the region, because they quantify, in monetary terms, the economic case for reallocating water from low-productivity agriculture to higher-productivity uses and for reducing Non-Revenue Water (USD 8.7 million in foregone revenue annually). The Mauritius accounts illustrate that institutional continuity is possible, with a five-edition series that tracks the UFW trend year-on-year. South Africa’s SWSA accounts map exactly which catchments are below ecological threshold, enabling targeted investment rather than blanket national programs.

7.2 Reflections on Data Limitations

Temporal misalignment. Country accounts cover different reference periods. For example, Uganda’s forest accounts run to 2015, while Zambia’s run to 2023, Kenya’s to 2024. Cross-country comparison is therefore partly comparing different economic and climate contexts.

Classification heterogeneity. Forest definitions, ecosystem typologies, and land cover classifications differ across countries and, in some cases, across accounting periods within the same country. For example, Ethiopia’s observed forest cover gain may partly reflect reclassification rather than genuine restoration. This limits quantitative cross-country comparison of forest change rates.

Physical-only accounts. For the majority of countries and ecosystem services, monetary valuation has not been completed. The analyses here use physical flows where monetary

values are absent. Converting these to monetary estimates would substantially strengthen the economic case for conservation investment but requires country-specific valuation studies.

Informal economy gaps. Charcoal, fuelwood, bushmeat, medicinal plants, and subsistence agricultural water use are systematically under-measured across the region. The DRC's 90 percent informal logging share and Uganda's 4.7 percent formal sector share of wood value are reminders that official accounts capture only a fraction of forest-dependent economic activity.

7.3 Recommendations

1. **Prioritize monetary ecosystem service accounts** for Ethiopia and Rwanda's ecosystem services, building on existing physical frameworks. Carbon sequestration valuation (using Article 6 carbon market prices) and water regulation services (using avoided cost methodologies) have the most established pathways and the highest policy demand.
2. **Support energy supply and use table development** in Ethiopia, the DRC, and Uganda as a priority for ASCENT program monitoring. Physical energy accounts are the minimum baseline; monetary energy supply and use tables (as developed in Kenya) provide the most policy-actionable insights.
3. **Invest in sub-national disaggregation** for water and land accounts in Kenya, Zambia, and the DRC. Basin-level and district-level accounts already exist in prototype form in several countries; systematic spatial disaggregation would transform national-level accounts into investment targeting tools.
4. **Institutionalize NCA within national statistical agencies.** The DRC, Ethiopia, and Zambia explicitly identify institutionalization as a priority. Technical assistance for placing SEEA compilation within the mandate and budget of national statistical offices (following the Mauritius and Uganda water account models) is the single most important intervention for ensuring that existing account investments generate long-term value.
5. **Develop cross-country harmonized accounts for shared resources.** The Zambezi Basin spans Zambia, Zimbabwe, and Mozambique. The Congo Basin spans the DRC, Republic of Congo, Cameroon, and others. Harmonized basin-level water and forest accounts would enable regional governance discussions and shared investment planning that national-level accounts cannot support alone.
6. **Initiate marine and coastal ecosystem accounts.** The AFE region's coastal economies depend heavily on marine resources that are entirely outside the current SEEA implementation footprint. South Africa's estuary accounts, which document sawfish extinction and near-total fish nursery degradation, illustrate what such accounts would reveal if compiled along the eastern African coastline.

7. **Use NCA data in ASCENT monitoring and evaluation.** Energy accounts (physical and monetary) provide the baseline against which ASCENT program outcomes (connections, clean cooking adoption, demand reduction) can be tracked in macroeconomic terms rather than only in count metrics. Countries with compiled energy accounts would benefit from integrating their ASCENT reporting into the NCA framework.

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